GIS

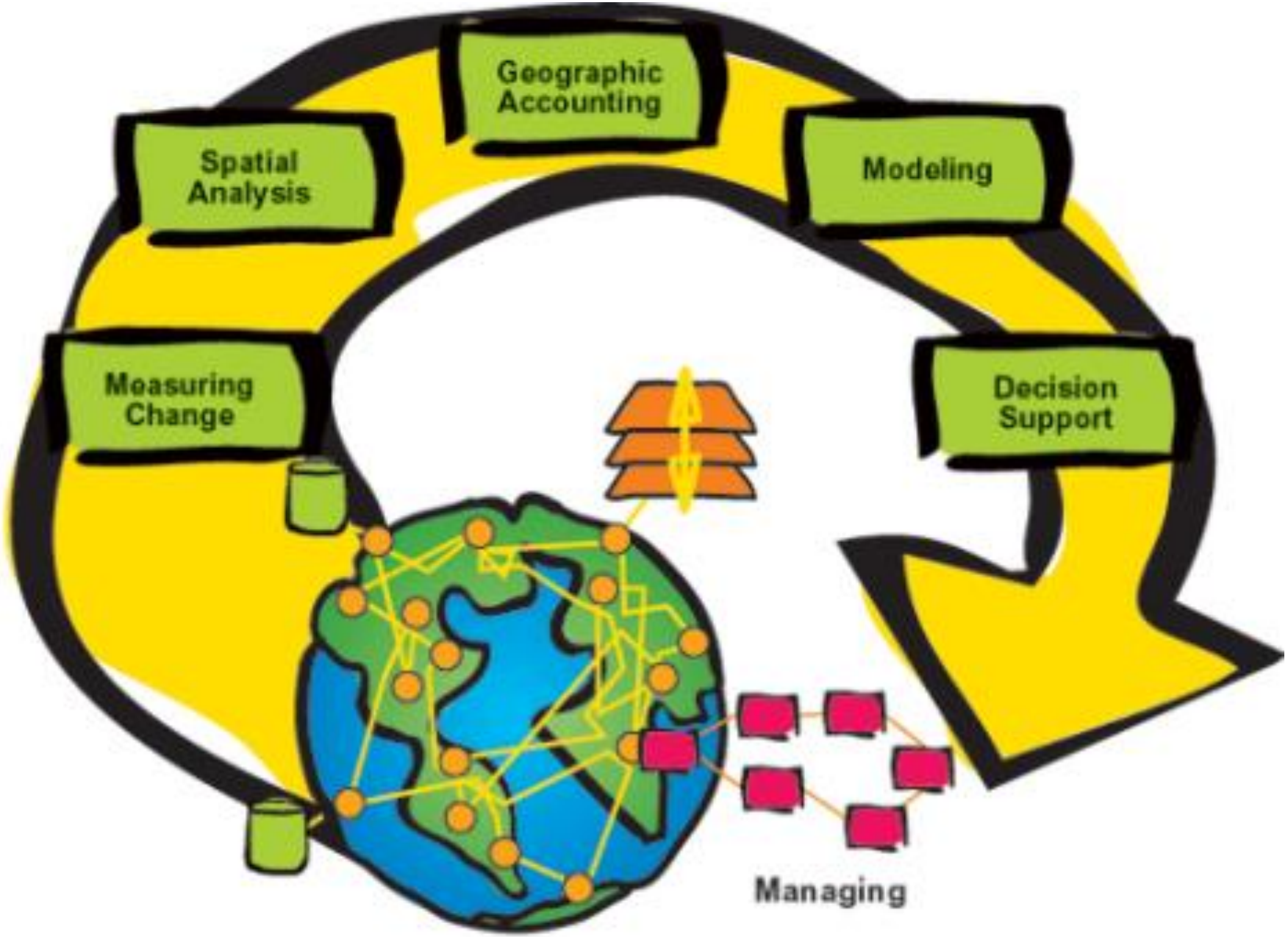
Using applied geography for understanding the real world and design a smarter future with strategic planning

WITH OUR GEOGRAPHIC INFORMATION SYSTEM, ANYTHING IS POSSIBLE.

Our tools help people see data in new ways to design solutions for business, government, and the environment.

The world is a big place made up of complex interactions.
Our platform helps make sense of it all.

A geographic information system (GIS) lets us visualize, question, analyze, and interpret data to understand relationships, patterns, and trends.


Geographic Accounting
Spatial Analysis
Modeling
Measuring Change
Decision Support
Managing

A Global GIS

- Uses of GIS range from indigenous people, communities, research institutions, environmental scientists, health organizations, land use planners, businesses, and government agencies at all levels.

- Uses range from information storage; spatial pattern identification; visual presentation of spatial relationships; remote sensing - all sometimes made available through internet web interfaces, involving large numbers of users, data collectors, specialists and/or community participants.

- Insurance
- Retail
- Manufacturing
- Real Estate
- Banking
- Media & Entertainment
- Supply Chain

- Federal, State, Local
- Defense
- Intelligence
- Resilient Communities
- Architecture, Engineering, and Construction (AEC)
- Economic Development
- Elections & Redistricting
- Facilities
- Land Administration
- Public Works
- Survey
- Urban & Regional Planning

- Agriculture
- Climate, Weather, and Atmosphere
- Conservation
- Environmental Management
- Forestry
- Mining
- Oceans
- Petroleum
- Water Resources

- Emergency Call-Taking, CAD, and RMS
- Emergency/Disaster Management
- Fire and Emergency Services
- Homeland/National Security
- Law Enforcement

- Airports & Aviation
- Highways
- Ports & Maritime
- Public Transport
- Railways

- Electric
- Gas
- Pipeline
- Telecommunications
- Water Utilities

- Education
- Health & Human Services
- Mapping | Statistics | Imagery
- Sustainable Development

GIS benefits organizations of all sizes and in almost every industry. There is a growing interest in and awareness of the economic and strategic value of GIS.

Purpose and Benefit of GIS

- **<u>Make Better Business Decisions</u>**
  By mixing regional and location-related information with other company information companies can gain critical ideas that help their businesses succeed. With spatial issues companies can select trading places, assistance areas and clients of interest, and existing the details in thematic maps and reviews **to accomplish faster and better company choices.**
- **<u>Improve Functional Performance & Reduce Cost</u>**
  The details provided by GIS is used to greatly improve operational preparing and control, and to re-engineer and improve company procedures. It gives you solutions for redirecting optimization, and servicing preparing and confirming.

- **Enhance Customer Service and Increase Sales**
  Understanding your clients, providing them with the best possible assistance and finding more people like them is central to increasing revenue. GIS helps organization to recognize their best clients and use spatial issues to determine regional areas where similar census apply in order to focus on industry development.
- **Better & More Cost-effective Plan Citizens**
  Government companies, especially municipality, face remarkable difficulties in the economy. They have to face providing increasing communities, with higher objectives of assistance, while dealing with reducing sources. GIS provides an efficient remedy to improving key resident services such as permit/licensing, preparing, servicing, features control and trains and buses.

Decision making becomes better

- GIS is known for go to technology, for it makes better decision about area and location. Making right decision about certain location is hard and critical. That is why GIS is used to help and give an idea to formulate better decision to implement.

GIS Functions

CAPTURE
- GPS satellites

STORE
- Stores in Vector or Raster format
- Spatial and Attribute data

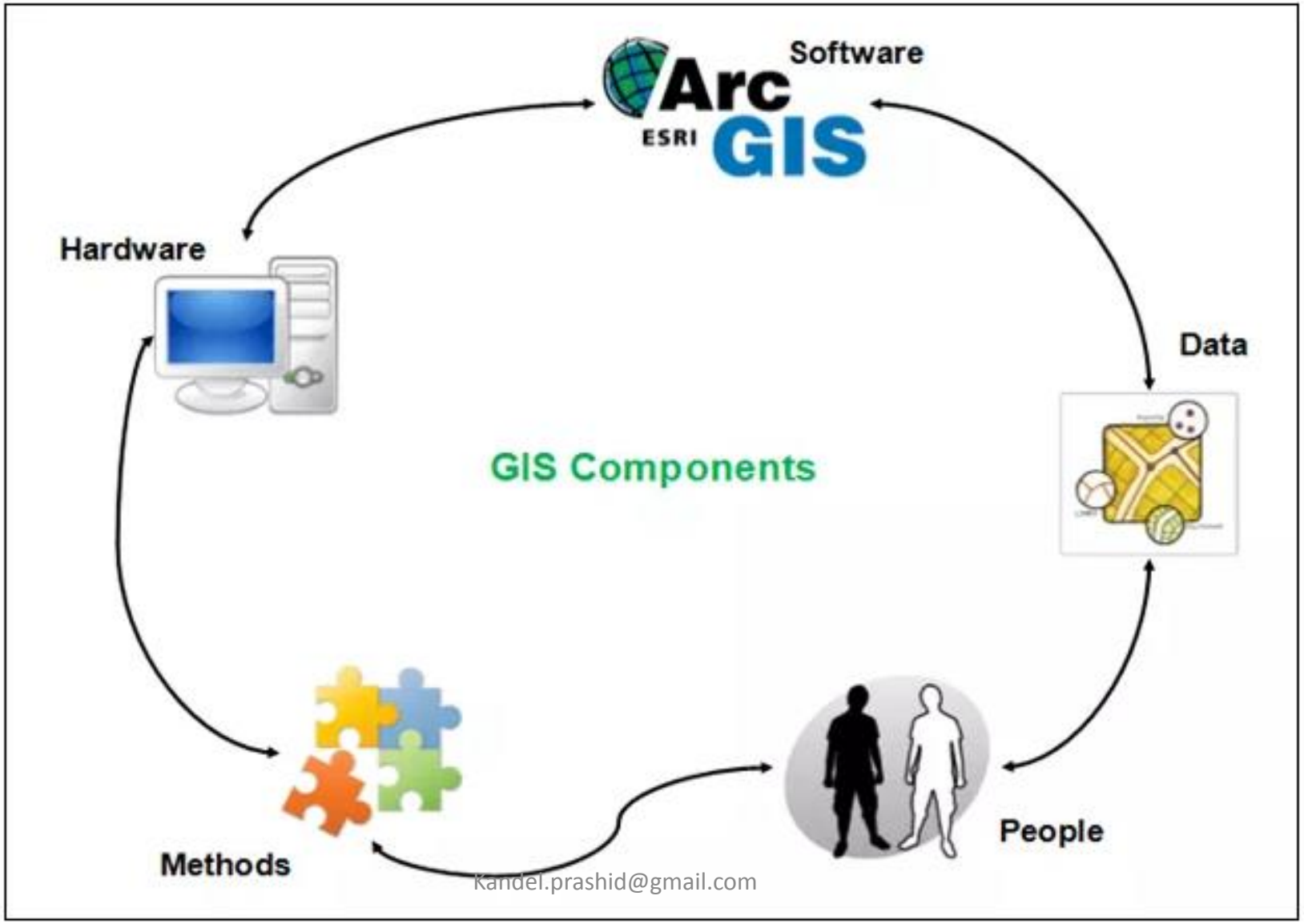
QUERY
- Identifying specific features
- Identifying features based on conditions

ANALYSIS
- Proximity analysis
- Overlay analysis-overlap 2 layers
- Network analysis-examines linear features

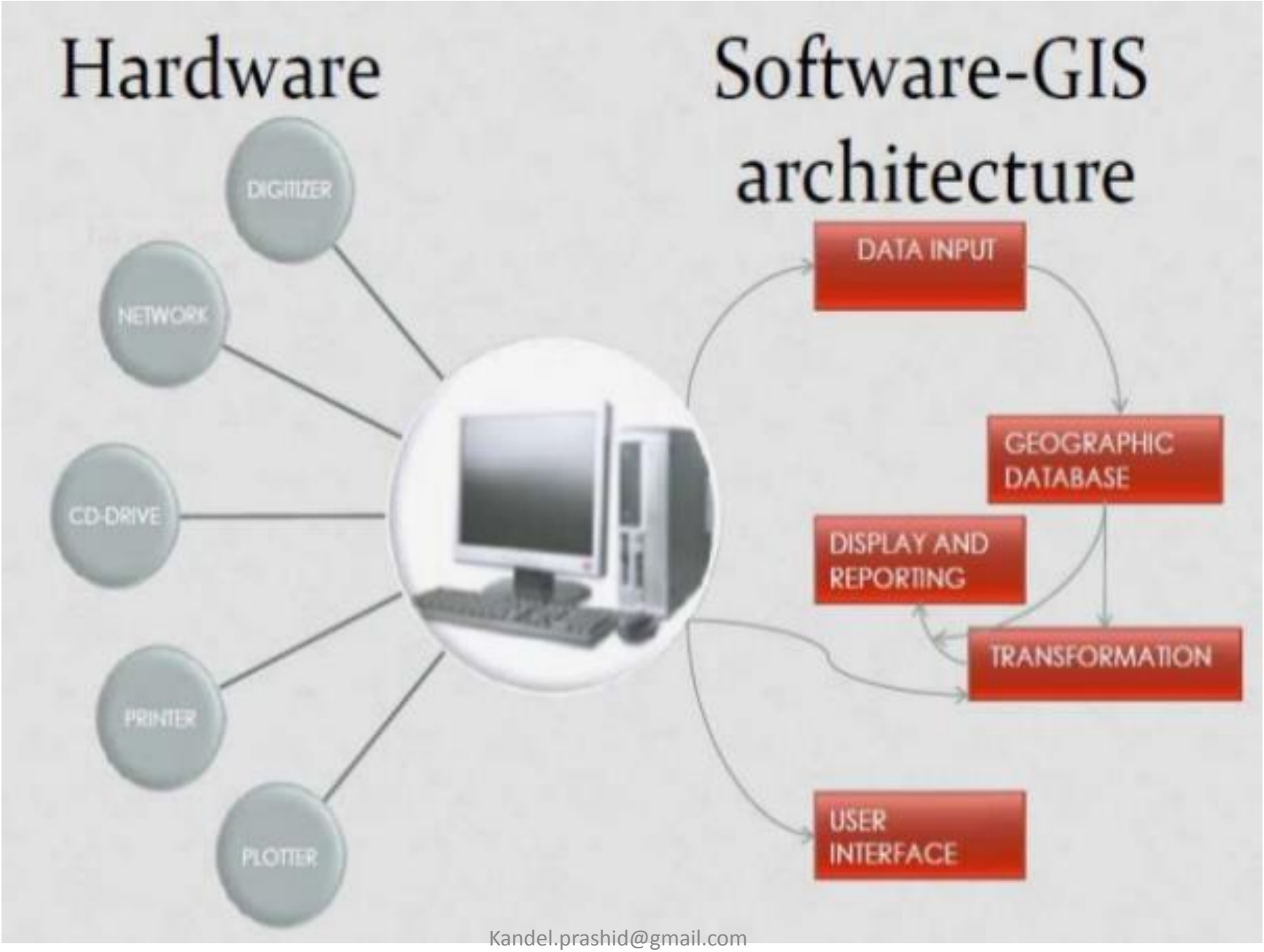
DISPLAY
- Visualized as a maps
- Reports, 3-D views, photographs

OUTPUT
- Paper map, Document, Image
- Internet

Components of GIS



A Geographic Information System (GIS) links locational (spatial) and database (tabular) information and enables a person to visualize patterns, relationships, and trends. This process gives an entirely new perspective to data analysis that cannot be seen in a table or list format. The five components of a GIS are listed below.


Hardware
Software-GIS architecture
DIGITIZER
NETWORK
CD-DRIVE
PRINTER
PLOTTER
DATA INPUT
GEOGRAPHIC DATABASE
DISPLAY AND REPORTING
TRANSFORMATION
USER INTERFACE

IMPORTANCE OF RS AND GPS in GIS

- In any GIS project, 80% time and money is spend in data acquisition.
- Remote Sensing and GPS are two important parts of GIS that helps to acquire dataset of real world.
- GPS assist to collect location data (x,y,z)as a basic and will be generated polyline and polygon form that multiple point.
- The Global Positioning System (GPS) is a space-based satellite navigation system that provides location and time information in all weather conditions, anywhere near on or near the Earth where there is an unobstructed line of sights to four or more GPS satellites.

GARMIN
Gasverksvej
10:39

Heart Rate
70
62

GARMIN
1:25.18
490 ft

0.2
Curve Ahead 0.2
Danny Thomas Blvd
55
GARMIN
Speed
52
I-40
Arrival
2:27

Remote Sensing

- While Remote Sensing also helps in grabbing the information of real world either by the satellite imageries or by the means of Aerial Photographs.
- *"Remote sensing is the science (and to some extent, art) of acquiring information about the Earth's surface without actually being in contact with it. This is done by sensing and recording reflected or emitted energy and processing, analyzing, and applying that information.“*

A – ENERGY SOURCE OR ILLUMINATION

B – RADIATION OF THE ATMOSPHERE

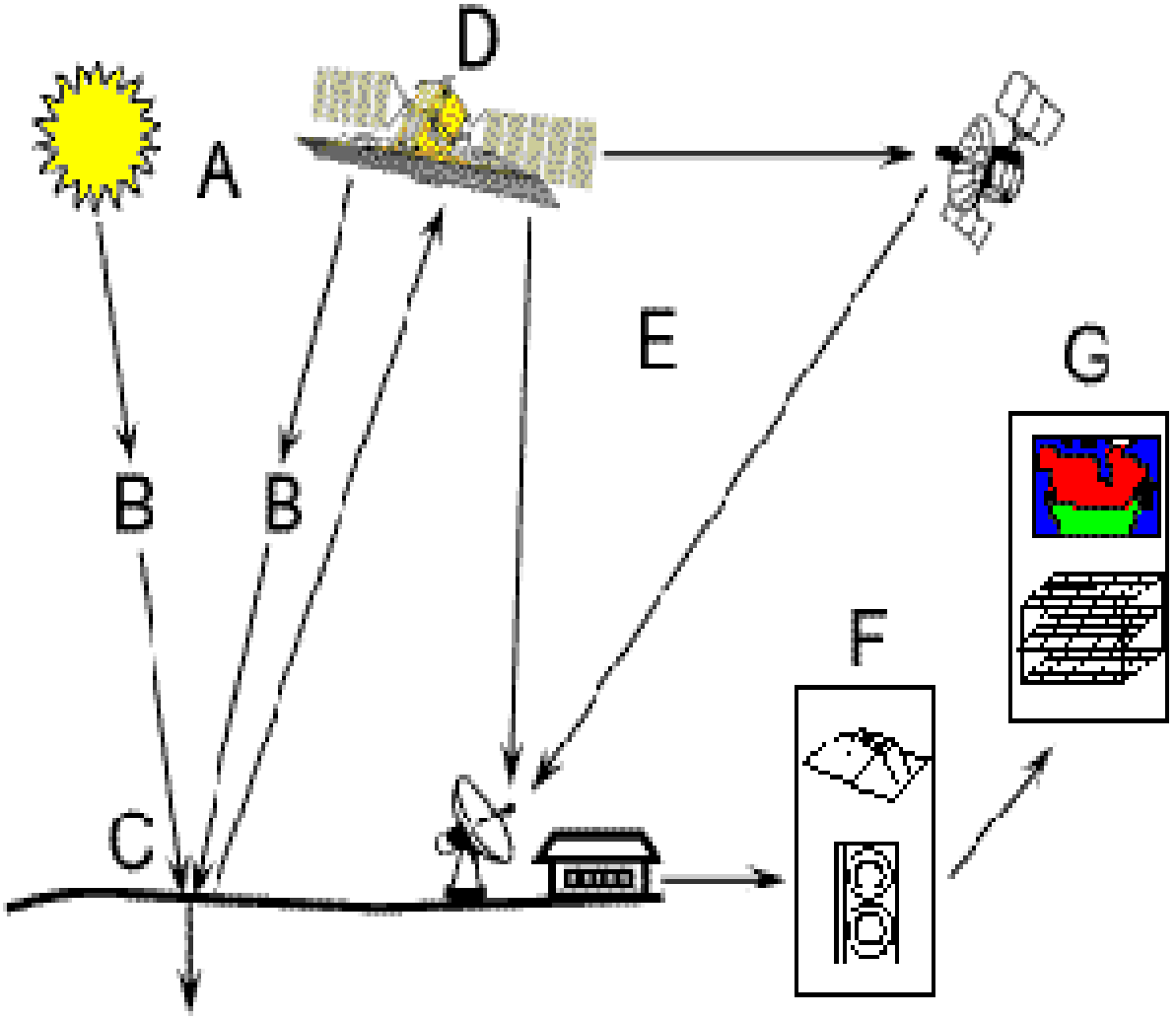
C - INTERACTION WITH THE TARGET

D – RECORDING OF ENERGY BY THE SECTOR

E – TRANSMISSION, RECEPTION, AND PROCESSING

F – INTERPRETATION ND ANALYSIS

G – APPLICATION



WorldView-3
WorldView-2
GeoEye-1
WorldView-1
WorldView-4

Honolulu
United States

Of our five senses (sight, hearing, taste, smell, touch), three may be considered forms of "remote sensing", where the source of information is at some distance. **The other two rely on direct contact with the source of information - which are they?**

THANK YOU!! UNIT 1 COMPLETED